

The Interplay Between Exercise Metabolism, Epigenetics, and Skeletal Muscle Remodeling

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SEABORNE, R.A. and A.P. SHARPLES. The interplay between exercise metabolism, epigenetics, and skeletal muscle remodeling. *Exerc. Sport Sci. Rev.*, Vol. 48, No. 4, pp. 188–200, 2020. We explore work from within the field of skeletal muscle and across the broader field of molecular biology, to propose that the link between exercise and skeletal muscle adaptation lies in the interplay between metabolism and epigenetics. Future investigations into such an interaction are crucial to advance our understanding of the beneficial effects of exercise on performance and health. **Key Words:** epigenetics, skeletal muscle, exercise, metabolism, DNA methylation, histone modifications

Key Points

- Exercise changes the abundance of a number of important metabolites and substrates to keep our muscles contracting.
- These metabolites may serve as substrates to enable epigenetic (biochemical, nongenetic modifications) modifications.
- These epigenetic modifications may, in turn, enable transcriptional changes to key signaling pathways and networks.
- Over a period of days, weeks, and months, this interaction can enable skeletal muscle adaptation, remodeling, and morphological changes that enhance health and performance.

INTRODUCTION

Skeletal muscle represents one of the largest constituents of the human organism, comprising ~30%–40% body weight of the mature adult phenotype. The integrity, health, and function of skeletal muscle is strongly associated with a host of diseases such as hypertension, type 2 diabetes (T2D), obesity, and age-associated muscle wasting (sarcopenia). Disorders in this tissue have significant repercussions on humans, with attenuated quality of life, health span, and earlier mortality. Exercise, alongside sensible dietary habits, remains the leading preventative

measure for skeletal muscle deterioration and has, for example, proven more successful than pharmaceutical interventions in the treatment of T2D and sarcopenia (1). Nonetheless, with rising sedentary behavior and adverse life/work-style regimes (e.g., office-based job), exercise participation rates are alarmingly low and in certain circumstances inaccessible (e.g., hospitalization). Therefore, there is great interest in understanding the molecular mechanisms by which exercise permits its beneficial effects on skeletal muscle, in the hope that we can harness this knowledge to help attenuate age-associated loss of muscle mass and function. In an elegant review article several years ago (2), Egan and Zierath (2) highlighted some of the underlying metabolic and molecular perturbations that occur after both acute and chronic exercise (aerobic and resistance) that permit functionally important adaptations at the whole-tissue level. Since this review, molecular physiologists have begun to unravel the role epigenetic modifications play in muscular adaptation, with a marked increase in recent publications. Given the increasing importance of understanding the molecular mechanisms that govern tissue and organism health, this field of work has important implications for both human performance and disease. It is, therefore, the purpose of this review to discuss the potential interplay between exercise metabolism and epigenetics and to integrate the biological phenomenon into our understanding of the molecular basis of skeletal muscle adaptation. We first describe the traditional metabolic program in skeletal muscle induced after aerobic and resistance exercise (RE). The review then introduces a new theoretical paradigm that is not yet established empirically, in which epigenetic modifications are elicited as a consequence of skeletal muscle metabolic flux. We draw upon classical biological understanding and data/conclusions from the wider field of metabolism and molecular biology to draw causal links between

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exercise-induced changes in metabolite, substrate, and cofactor availability, and the effects this has on key enzymes involved in DNA methylation and histone modifications as well as the potential consequences of these changes. To this end, we suggest that examining the interplay between metabolism, epigenetics, and skeletal muscle adaptation after exercise should be an important area of concerted and coordinated future research by physiologists and molecular biologists.

CURRENT UNDERSTANDING OF EXERCISE-INDUCED SKELETAL MUSCLE ADAPTATIONS

Our classical understanding of the molecular basis of skeletal muscle adaptation is underpinned by a series of sequential processes. Exercise, dependent on modality and intensity, creates an abundance of extracellular molecules (such as hormones, growth factors, and cytokines) secreted by various glands, tissues, and organs in the body. Through a number of mechanisms, these molecules and ligands ignite a hive of intracellular activity, including the change in activity of intracellular proteins, the binding/displacement of transcription factors, and the translocation of key molecules. These subsequently leads to the attenuation/augmentation in expression of key gene networks, as well as key posttranslation modifications, creating a coordinated and time-dependent response to an exercise stress. This process has been elegantly reviewed elsewhere (2). After both RE and aerobic exercise, gene expression changes occur in a transient manner, observably increasing approximately 2–4 h after the stimulus before returning to basal levels approximately 20–24 h later (3). Such transient changes in mRNA expression are a crucial hallmark of later adaptation, with repeated changes in expression (typically but not exclusively) leading to a complementary increase/decrease in the abundance of the coding protein produced by the ribosome (4). Ultimately, exercise-induced skeletal muscle adaptation is determined by the change in the relative abundance of functional proteins, in an attempt to maintain tissue homeostasis during continued or repeat encounters with the same/similar insult (4).

Transcriptional outputs rely on accessibility of fundamental transcriptional apparatus to the gene loci of interest. It is, therefore, important in the context of postexercise adaptation to have a functionally permissive chromatin, in which transcription factors and polymerase machinery are able to recognize, bind, and transcribe the DNA sequence. Epigenetic modifications converge upon regulatory sites to modify the conformational structure of the chromatin, generating either functionally active/permissive or inactive/repressive chromatin. Importantly, epigenetic modifications themselves are dependent on a host of substrates, cofactors, and metabolites that are generated in cellular energy pathways, thus linking a distinct interplay between cellular metabolism and epigenetics. Given that exercise stress is a potent modifier of energy and metabolic flux, the interplay between exercise metabolism and epigenetic modifications should be clear. Furthermore, it is, therefore, clear that this essential biological mechanism could be paramount in facilitating post-exercise adaptation and allowing the remodeling of skeletal muscle tissue. However, experimental work identifying the role, behavior, and function of epigenetics in this regard has only recently begun to gain attention.

This review focuses on how exercise changes the metabolic and molecular programs within the cell, which lead to modifications

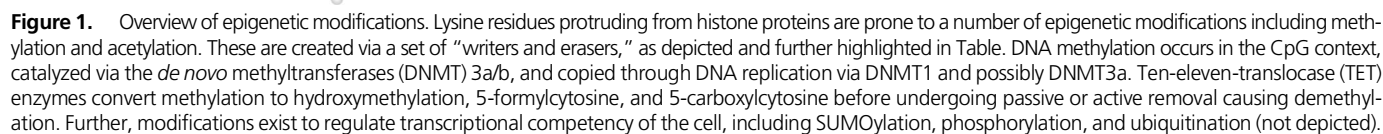
at the epigenetic level regulating the postexercise remodeling of skeletal muscle. It is beyond the scope of such a review to comment on all known epigenetic modifications; thus, we focus primarily on DNA methylation and histone modifications, given they are the best described and have a fundamental link between metabolic and molecular programs in other cell types and physiological models.

BRIEF OVERVIEW OF EPIGENETIC MODIFICATIONS

Epigenetics as a concept was first conceived by Conrad Waddington in 1940. Since then, the field has seen a rapid and exponential increase in attention following the turn of the century. Despite this, epigenetics remains an ambiguous term with multiple working definitions currently used. Herein, we use the definition of epigenetics, as changes in transcriptional expression/activity that are not due to underlying variation in genetic sequence. These modifications are of crucial importance in fundamental biology, regulating such processes as X chromosome inactivation and genomic imprinting (5), as well as being the most reliable molecular method of human biological age prediction (6), collectively underlying its importance and association with development, disease, and aging. Underlying epigenetic modifications permit genetically identical cells to achieve extreme divergence in phenotypic outcome, regulating the transcriptional program evoked during homeostasis and after encounters with stress. To date, hundreds of epigenetic modifications have been identified, the most prominent of which occur at two key regulatory sites within the genome. The underlying mechanisms by which human skeletal muscle displays extreme variation in its ability to adapt to a given exercise stress are yet to be fully elucidated; however, it is clear epigenetic modifications have an important role in physiological adaptation.

Histone Modifications

Histone proteins form an octamer structure (nucleosome) that tightly binds genomic DNA into a chromatin DNA-protein complex, condensing our genetic material into a nucleus (Fig. 1). Protruding from these histone proteins are N-terminal amino acid tails that are accessible for a host of epigenetic modifications, the most common are: methylation, SUMOylation, acetylation, phosphorylation, and ubiquitination. The consequential effect of these modifications is dependent on the histone protein, amino acid residue, and the type and level of modification. For example, trimethylation of lysine 27 of histone 3 (H3K27me3) is associated with transcriptional repression via the formation of heterochromatin regions (7). However, dimethylation of arginine 17 on the same histone protein (H3R17me2) is associated with euchromatin and permissive transcriptional consequences (8). Further examples include those observed on the same mark but with differing levels of modification, where H3K27me1 is associated with active gene promoters, but the same modification mark when trimethylated is functionally repressive. Important regulatory enzymes have also been identified to initiate the induction, reduction, and maintenance of histone variant marks, aptly named as writers, erasers, and readers, respectively. We guide the reader to previous extensive reviews on these regulatory enzymes (9,10), as well as our depiction and description of these in Figure 1 and Table. For example, both the SET domain protein family and the mixed lineage leukemia (MLL) family are responsible for a vast majority of histone methyltransferase



acetylation, and deamination, but are beyond the scope of this review.

Wrapped around the histone octamer is 147 base pairs (bp) of DNA, with up to 80 bp of linker DNA separating each core nucleosome particle. In mammals, DNA is prone to single-level DNA methylation modifications in a CpG context, where methyl molecules are removed or added to the fifth carbon position of a cytosine (C) nucleotide (Fig. 1). Much of the human genome is sparsely populated by CpG sites that are heavily methylated, with brief intervening regions of highly dense CpG regions that are lowly methylated. These regions are known as CpG islands and commonly associate with gene bodies, including promotor regions (14). Located up- and downstream of CpG islands are CpG shores and CpG shelves, which are less dense and display greater tissue-specific methylation profiles (15). Hypomethylation occurs via a number of methyltransferase enzymes (DNMTs), with DNMT3a and 3b acting in a *de novo* fashion to establish DNA hypermethylation, whereas DNMT1 largely acts upon hemimethylated CpG sites during DNA replication (Fig. 1), installing previous modifications and enabling the maintenance of methylation marks across cellular

TABLE. Summary of key enzymes associated with writing, reading, and erasing methylation and acetylation marks on both DNA and histone lysine residues

	Target	Writer		Reader	Eraser	
	Site	Enzyme	Level of Modification	Enzyme/Proteins	Enzyme	Level of Modification
DNA methylation	5-mC DNA	DNMT3a	—	MeCP2	TETs	—
	5-mC DNA	DNMT3b	—	MBD1–6	TETs	—
	5-mC DNA	—	—	SETDB1/2	TETs	—
	5-mC DNA	—	—	ZBTB33/Kaiso	TETs	—
	5-mC DNA	—	—	ZBTB4	TETs	—
	5-mC DNA	—	—	ZBT38	TETs	—
	5-mC DNA	—	—	ZFB57	TETs	—
Histone lysine methylation	Hemimethylated DNA	—	—	DNMT1 + UHRF1	Passive	—
	H3K4	SET1 a/b	me1, me2, me3	BPTF	KDM1a, b	me1, me2
	H3K4	MLL1/2	me1, me2, me3	TAF3	NO66	me1, me3
	H3K4	MLL3/4	me1, me2	ING4	KDM5a, c, d	me2, me3
	H3K4	SMYD1–2	me1	SPIN1	KDM5b	me1, me2, me3
	H3K4	SET7/9	me1	SGF29	—	—
	H3K4	PRMD9	me1, me2, me3	CHD1	—	—
	H3K9	SUV39H1/2	me2, me3	HP1 α	KDM1a	me1, me2
	H3K9	G9A	me1, me2	HP1 β	KDM4a–c	me2, me3
	H3K9	GLP	me1, me2	HP1 γ	KDM4D	me1, me2
	H3K9	SETDB1	me1, me2	UHRF1	PHF8 family	me1, me2
	H3K9	—	—	—	KDM3a–c	me1, me2
	H3K27	EZH1	me2, me3	CBX7	UTX/UTY	me2, me3
	H3K27	EZH2	me2, me3	EDD	JMJD3	me2, me3
	H3K27	—	—	BAHD1	KIAA1718	me2
	H3K27	—	—	NSD2	PHF8	me2
	H3K36	SETD2	me3	DNMT3a/l	KDM2 family	me1, me2
	H3K36	NSD1–3	me1, me2	LEDGF	K4a–c	me2, me3
	H3K36	SETD3	me1, me2	NSB1	KDM8	me2
	H3K36	SMYD2	me2	KU70	—	—
	H3K36	ASH1L	me2	MRG15	—	—
	H3K36	SETMAR	me2	ZMYND11	—	—
	H3K79	DOT1L	me1, me2, me3	TP53BP1	KDM2b	me2, me3
	H4K20	SET8	me1	ORC1	PHF8	me1
	H4K20	SUV4-20H1/2	me2, me3	Sir3	PHF2	me1, me3
Histone lysine acetylation	H3K9	GCN5	—	BRD4, 7	SIRT1, 2, 5, 6	—
	H3K9	GTF3C4	—	TAF1	HDAC11	—
	H3K9	MYST3	—	PBAF	—	—
	H3K9	GTF3C4	—	PCAF	—	—
	H3K14	GCN5	—	ATAD2	SIRT1–3	—
	H3K14	P300	—	BAZ2b	HDAC11	—
	H3K14	MYST3	—	BRD2, 7	—	—
	H3K14	—	—	PBAF	—	—
	H4K8	P300	—	BRD7	HDAC3 (?)	—
	H4K8	TIP60	—	BRDT	?	—
	H4K8	ELP3	—	GCN5	—	—
	H4K8	MYST2	—	PCAF	—	—
	H4K8	MYST3	—	BRG1	—	—
	H4K8	—	—	ZMYND8	—	—
	H4K12	MYST2	—	BRD2–4, 7	HDAC3	—
	H4K12	MYST3	—	BRDT	?	—
	H4K12	TIP60	—	BRDF1	—	—
	H4K12	HAT1	—	P300	—	—
	H4K12	—	—	BRG1	—	—
	H4K16	MOF	—	BRD2–4, 7	SIRT1, 2, 3	—
	H4K16	TIP60	—	BRDT	—	—
	H4K16	MYST3	—	BPTF	—	—
	H4K16	MYST4	—	ZMYND8	—	—
	H4K16	GCN5	—	GCN5	—	—

divisions. Reversal of DNA methylation marks (demethylation/hypomethylation) can occur in either a passive or an active manner. Passive hypomethylation is thought to occur during DNA replication, whereby in the absence of DNMT1 binding, 5-hydroxymethylcytosine (5-hmC) are unrecognized and the replicating strand of DNA acts like an unmodified cytosine; in essence, methylation is “diluted”/passively lost from DNA (16). In active hypomethylation, the family of Ten-eleven translocase (TET) enzymes initiate the reduction of 5-methylcytosine (5-mC) to multiple formats (e.g., 5-hmC, 5-fC) through deamination or oxidization, before the modified nucleotide is highlighted for removal and replacement, by base excision repair (17). Because of the requirement of TETs, active removal of DNA methylation is regulated by a number of elements, most notably by metabolic substrate and cofactor availability (17), highlighting an important interplay between metabolism and epigenetics.

METABOLITES AND EPIGENETIC REMODELING

The activity and behavior of the most influential DNA and chromatin modifiers are due to the availability and concentration of metabolites and cofactors, including acetyl coenzyme A (acetyl-CoA), adenosine triphosphate (ATP), alpha-ketoglutarate (α -KG), S-adenosylmethionine (SAM), nicotinamide adenine dinucleotide (NAD⁺) amongst many more. The availability and concentration of these metabolites and substrates are largely due to the metabolic state of the cell, where nutrient availability and cellular stress are likely modulators of this response. The next section focuses on the interplay between cellular metabolism and epigenetics. Because of space limitations, we focus primarily on metabolite associations with DNA methylation, histone methylation, and acetylation modifications. A number of previous review articles have elegantly reported the link between metabolism and epigenetics and should also be explored (18–20).

Histone Acetylation

Acetylation occurs across a number of lysine residue sites in the histone complex, the most prominent of which are (H3K9, H4K5, H4K8) known to be transcriptional activators. Indeed, lysine acetylation decreases the electrical charge between histones and negatively charged DNA, thereby opposing their association and permitting an open and transcriptionally active state (21). The enzymes responsible for increased lysine acetylation (e.g., writers) can be categorized into three families: histone acetyltransferases (HATs) of p300, GCN5, and MYST. Once established, histone acetylation marks act as anchor sites for the recruitment of bromodomain-containing proteins that themselves act as readers and modifiers to regulate gene expression. HAT enzymes utilize the substrate acetyl-CoA, donating the acetyl group to relevant lysine histone marks and depositing CoA, where both the absolute level of acetyl-CoA and the ratio of acetyl-CoA to CoA have been shown to regulate lysine acetylation (22). The enzyme ATP-citrate lyase (ACLY) enables the conversion of citrate and CoA to produce nuclear acetyl-CoA in an ATP-dependent process. The availability of citrate is dependent on a key regulatory step in the tricarboxylic acid (TCA) cycle, where citrate synthase converts oxaloacetate into citrate via the use of pyruvate-derived acetyl-CoA. This reaction in the TCA cycle renders CoA (from the acetyl-CoA) and available citrate, which is either further metabolized in the

TCA cycle or shuttled to the cell nucleus, where it is available for the aforementioned reaction. Importantly, it has been shown that nuclear availability of acetyl-CoA is pivotal for the modification of lysine acetylation, compared with cytoplasmic availability. In extensive *in vitro* work in human and mouse cell lines (23), it has been suggested that TCA cycle-derived acetyl-CoA is the critical substrate for modification of lysine acetylation and, thus, indicates a key relation between cellular energy status, TCA activity, and epigenetic behavior (23). Such metabolic-epigenetic interplays are important for global transcriptional programs but notably, augment the transcriptional profile of genes associated with glycolysis efficiency (e.g., glucose transporter 4) (23) and resistance to oxidative stress (e.g., Forkhead box O3a) (24). Thus, the associations between metabolism, epigenetics and tissue adaptation are clear but as yet, are not studied in an exercise- or skeletal muscle-related context.

Histone Deacetylation

The coenzyme NAD⁺ becomes an import cofactor when lysine deacetylation is observed. In a converse manner to that identified above, histone deacetylases (HDACs) induce opposing electrical charges between histones and DNA, thus promoting the association of the two genomic loci and establishing the heterochromatin and a transcriptionally repressed state. HDACs, as removers of acetylated marks, can be categorized into two metabolite-dependent groups: the zinc-dependent class (HDACs I, II, and IV) and the NAD⁺-dependent class (HDACs III). The latter, more commonly referred to as sirtuins (SIRT1), utilize NAD⁺ to deacetylate lysine residues, thus modulating gene accessibility and transcriptional potential but, importantly, in a targeted manner. For example, SIRT1 specifically targets H3K9ac and K3K14ac marks, whereas SIRT6, while targeting H3K9ac, also independently targets H3K56ac. NAD⁺ is an oxidizing substrate and is accepting of new electrons, whereas its reduced form (NADH) donates electrons, with this process being crucial in the electron transport program in skeletal muscle metabolism. Crucially, when there is a surplus supply of energy, the amount of NAD⁺ to NADH is reduced, as evidenced during high glycolytic activity. Subsequently, such plentiful supply of energy is detrimental to the HDAC activity of SIRT elements, rendering hyperacetylated lysine residues and transcriptionally active chromatin. However, during states of low-cellular energy status, the ratio of NAD⁺/NADH is permissive of SIRT activity, and thus, HDAC modifications are enabled, for which exercise-induced skeletal muscle adaptations are artifacts. For example, SIRT1 has been shown to physically bind to the peroxisome proliferator activated receptor gamma coactivator 1 (PGC1 α) protein, a key metabolic gene, resulting in SIRT1 functionally modifying the acetylation profile of PGC1 α (25). Counterintuitively, reduced acetylation of PGC1 α alters its nuclear localization, allowing it to become more transcriptionally active; thus, increased SIRT1 binding/modification increases PGC1 α expression. PGC1 α expression is considered a hallmark of aerobic exercise adaptation and is classically known to induce a myriad of responses in skeletal muscle (including mitochondrial DNA synthesis, mitochondrial biogenesis and enhanced oxidative metabolism) (26), that allow the tissue to adapt enabling enhanced exercise performance/capacity. Deacetylation of PGC1 α

via SIRT1 is commonly suggested to be under the regulatory control of the adenosine monophosphate complex, AMP-activated kinase (AMPK) (27). In an endurance exercise perspective however, SIRT1 knockout (KO) models identified a different approach in the regulation of PGC1 α acetylation levels. Indeed, the acetyltransferase GCN5 was shown to decrease in activity after SIRT1 KO with a corresponding reduction in the acetylation profile of PGC1 α (28). Nonetheless, the work detailed here clearly suggests a prominent interplay between metabolism, epigenetics, and skeletal muscle adaptation.

Histone and DNA Methylation

As previously mentioned, hypermethylation of both histone and DNA loci revolves around the transfer and addition of a methyl molecule at both locations. A divergent set of enzymes regulate such a *de novo* methylation mark at CpG and lysine residues, where DNMT1, 3a, and 3b enzymes modulate CpG methylation, and enzymes such as SUV39H1, EZH1 and SETD2 regulate H3K4me₃, H3K27me₃, and H3K36me₃, respectively. Nonetheless, these processes require the same metabolite, SAM, that allows the donation of the methyl molecule, rendering a methylated mark and its reduction to the molecule S-adenosylhomocysteine (SAH). The bioavailability of SAM is crucial for all forms of methylation, as indicated by studies where cells grown under conditions of methionine restriction (a precursor to SAM production) display a decrease in H3K4me₃, and a subsequent effect on transcriptional potential of targeted genes (29). The synthesis of SAM occurs predominantly through methionine metabolism in the methionine cycle, but it is strongly associated with the accompanying folate cycle (Fig. 2). Through methionine adenosyl transferase (MAT) enzymes and the use of ATP, methionine is converted into SAM, which is subsequently broken down into SAH after donation of a methyl molecule in methylation reactions. SAH is a potent inhibitor of MAT activity (30), and thus is rapidly broken down into its constituent parts of adenine and homocysteine by SAH hydrolase (SAHH). Homocysteine is recycled into methionine through a cross-talk with the folate cycle and one-carbon metabolism. Interestingly, the activity potential of SAHH is dependent on the availability of the cofactor NAD⁺ (31), thus suggesting that this cosubstrate has a dual role in regulating both histone and DNA hypermethylation, as well acetylation levels via HDACs (mentioned above). Because of the tandem link between one-carbon metabolism and the methionine metabolism cycle, a role for amino acid- and phospholipid-derived molecules has been identified. First, in the glycolytic pathway, phosphoglycerate dehydrogenase oxidizes the glycolysis substrate 3-phosphoglycerate to create serine (Fig. 2). Serine is one of several amino acids that serve the one-carbon folate pool via tetrahydrofolate (THF) and actively facilitate methionine synthesis from the precursor homocysteine (30), thus identifying an interaction between metabolism and epigenetic pathways. Furthermore, amino acids such as threonine have been shown to be vital for the regulation of methylation. Threonine modulates di- and trimethylation of H3K4 via the enzyme threonine dehydrogenase (TDH), which is evidenced via depletion of the amino acid *in vitro*, that causes a significant reduction in methylation that subsequently affects cell proliferation. In the same step of the methionine synthesis reaction, the phospholipid derivative of choline, betaine, also plays a crucial role in the balance of the ratio between SAM

and SAH. Indeed, when in abundance, the betaine homocysteine S-methyltransferase (BHMT) enzymes convert betaine and homocysteine into dimethylglycine and methionine, thus acting as a catalyst for the recycling of the SAM metabolism pathway (Fig. 2). In the context of hypomethylation at the lysine and DNA loci, it is clearly observed that metabolic products serve to enhance the activity and efficiency of key enzymatic processes and thus, clearly links metabolism and epigenetic modifications.

Histone and DNA Demethylation

Histone and DNA demethylases are orchestrated via a different set of enzymes. The former is controlled by a number of enzymes belonging to the oxidative lysine-specific histone demethylases (LSD or KDM) or the iron- and α -KG-dependent JmjC. Interestingly, LSD1 has high specificity for the disruption of methylation marks at two specific histone residues. Indeed, landmark work first identified that LSD1A targeted and demethylated, H3K4 mono- and dimethylation marks only, and was incapable of demethylating H3K4me₃ (32). It was later identified that the same demethylase enzyme was capable of recognizing and modifying H3K9me_{1/2}, shortly before a second homolog was identified (LSD1B) that was only capable of recognizing mono- and dimethylated marks on H3K4. Both LSD1A and B are flavin-dependent enzymes and require flavin adenine dinucleotide (FAD) as a cofactor. FAD is synthesized in the cytoplasm and mitochondria from the vitamin riboflavin, and is required for multiple processes to ensure cellular metabolic efficiency. A nuclear pool of FAD-synthesizing and FAD-hydrolyzing enzymes exist in the cell nucleus that creates a dynamic ratio of FAD production and reduction, respectively, producing FADH₂. The ratio between FAD and FADH₂ is an important biosensor for histone demethylase activity, given that LSD1A/B requires available FAD as a cofactor. In the mitochondria, during the succinate oxidation step of the TCA cycle, turning succinate into fumarate, electrons are transferred to mitochondrial available FAD, subsequently causing a reduction to FADH₂. Thus, disrupting the ratio of FAD and FADH₂ (Fig. 2). The importance of such a ratio has been identified in adipocytes, where under restriction of FAD availability, LSD1 is downregulated, rendering hypermethylated H3K4 marks. These modifications coincide with augmented transcription of genes associated with metabolic function, such as PGC1 α and pyruvate dehydrogenase 4 (PDK4). However, the importance in subcellular derived FAD/FADH₂ substrates (e.g., nuclear vs mitochondrial) and their role in LSD1-induced demethylation require further attention in skeletal muscle.

The ability to recognize, target, and disrupt methylation marks in a specific manner is conserved to the second, and largest, group of demethylases. The JmjC domain-containing proteins that are compartmentalized into families including KDM2, 3, 4, 5, and 6, which all display the ability to modulate specific histone marks. For example, KDM2, known as FBXL, consists of two homologs (A and B) that both specifically function to demethylate H3K36me₂, with KDM2B also able to recognize H3K3me₃. Whereas, the third subfamily, KDM3a/b, is responsible for the removal of mono- and dimethylation of H3K9. Interestingly, only the JmjC domain-containing histone demethylases conserve the ability to recognize, bind, and disrupt trimethylation of any known histone lysine mark (33). Collectively, these KDM

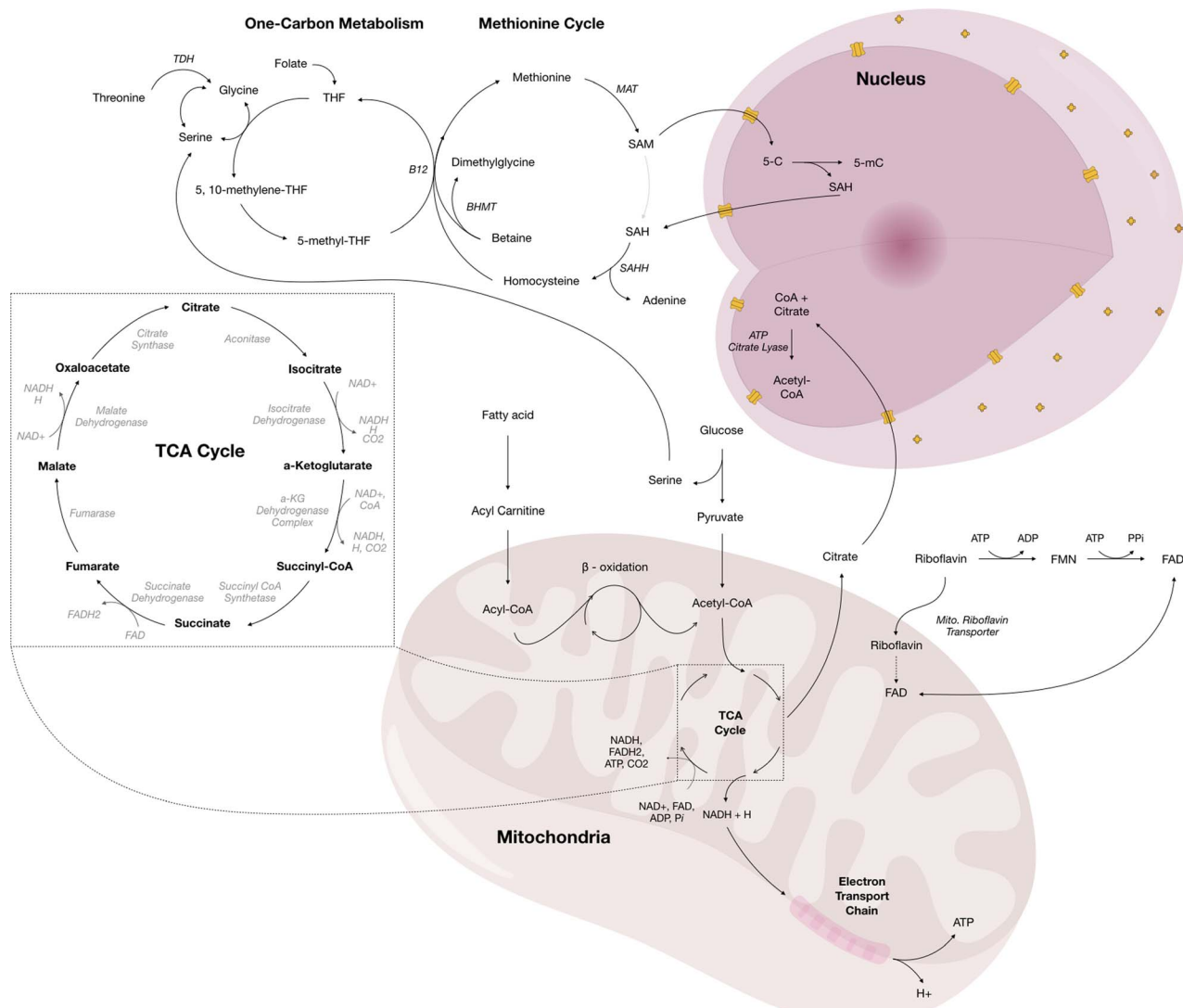


Figure 2. Simplified overview of metabolism, important intermediates, and the interplay to epigenetic modifications. Central to metabolism is the tricarboxylic acid (TCA) cycle, in which metabolites derived from nutrients (FAs and glucose) are broken down through sequential steps, producing a number of intermediary molecules, cofactors, and substrates. Key steps in this process are the production of α-ketoglutarate (α-KG), succinate, and fumarate that serve to regulate lysine and DNA demethylation activity.

enzymes differ from the LSD class of enzymes by the requirement of iron and α-KG as cofactors. In a similar fashion, the DNA-targeting demethylases, the TET enzymes, also show a dependency on α-KG to function. These class of dioxygenases catalyze the conversion of 5-mC to 5-hmC before 5-hmC undergoes one of two possible further mechanisms: either a passive loss of the mark during DNA replication or the active processing of the mark by demethylases that produces 5-formylcytosine and 5-carboxylcytosine, which is then removed via base excision repair. The process of methylation disruption from both DNA and lysine marks is similar, both processes utilizing the TCA cycle substrate α-KG and the cofactor iron. In the TCA cycle, α-KG is created via the oxidative decarboxylation of isocitrate by the enzyme isocitrate dehydrogenase (IDH) (Fig. 2). Three known homologs of IDH exist, with IDH2 and 3 responsible for the TCA cycle production of α-KG, whereas IDH1 catalyzes α-KG production in a cytoplasmic context, at which point the amino acid glutamine is a known rate limiting metabolite. Importantly, several TCA cycle steps downstream of the production of α-KG, succinate

and fumarate are produced, which act as intermediate inhibitors of both TET and KDM demethylase activity (Fig. 2), acting as a regulatory feedback mechanism. Given that FAD/FADH₂ ratio, production of α-KG, succinate, and fumarate are all dependent on the TCA cycle, demethylase activity is highly dependent on the flux of the TCA cycle. Herein exists one of the greatest examples of the interplay between metabolism and epigenetics.

Therefore, the interconnections between cell metabolism, metabolite, cofactor and substrate availability, and epigenetic modifications are clear. Therefore, it is of significant interest to molecular physiologists to understand how certain stressors manipulate the synthesis, transport, availability, and reduction of such metabolites. To this end, exercise, both aerobic and RE, provides a bolus stimulus in which metabolite production is rapidly modified to meet the demands of the exercised tissue. The next section focuses on how exercise modifies the metabolic control of skeletal muscle cells, with particular interest on the cofactors and substrates previously identified to be crucial in epigenetic enzymatic behavior.

INTERPLAY BETWEEN EXERCISE METABOLISM AND EPIGENETICS

ATP is integral for the successful contraction of mature skeletal muscle. The metabolite instigates the binding of the myosin head to available actin binding sites within the sarcomere unit, and once broken down, the resulting phosphate molecule initiates the myosin head to tilt and slide along the actin myofilament, creating the classically observed *sliding filaments*. It is, therefore, clear that to maintain adequate contractility of the skeletal muscle unit, ATP bioavailability becomes a rate limiting factor. To appease this high demand of energy, skeletal muscle rapidly metabolizes a number of available nutrients, including fats, carbohydrates, and amino acids, as well as recycling previously believed by-products such as lactate. During the up-regulation of such programs, previously highlighted metabolites and substrates such as acetyl-CoA, ATP, α -KG are produced in large amounts. Importantly, there are many regulatory elements that play a pivotal role in regulating skeletal muscle contractility and energy production, none more so than calcium (Ca^{2+}), which alleviates the troponin-tropomyosin complex and allows highly charged myosin heads to bind to actin binding sites. The following section reviews the effect of both aerobic and resistance exercise on energy production, metabolic pathways, the regulatory elements within these programs, and the interplay these may have on epigenetic modifications.

Glycolysis, One-Carbon Metabolism, and Exercise

Despite recent contention, it is affirmed that carbohydrate metabolism remains the paramount energy source during exercise, where intensities reach those of high to maximal exertion. Seminal work in the early 1990s reported a clear predominance for working skeletal muscles to favor the breakdown of muscle glycogen stores over muscular triglyceride reserves and, further, the metabolism of these products and plasma-derived glucose sources at aerobic exercise intensities of 85% $\dot{V}\text{O}_2\text{max}$ (34). There is clear association for the up-regulation of these metabolic programs during periods of incrementally increasing aerobic exercise intensity, as shown by significant increases in plasma glucose concentration (34), increased expression of the key glucose transport protein GLUT4, and a significant rise in the level of glucose uptake into skeletal muscle (35). During resistance exercise, the requirement for glycogen as a supporting energy source for muscular contraction is less well documented. Work in the 1990s suggests that there is a marked and noticeable reduction in muscle glycogen content after resistance exercise when performed at high (leg press at 70% 1 repetition maximum (1RM) for 6 repetitions) and lower (35% 1RM for >14 repetitions) exercise intensities (36). Further work confirming the reliance on glycogen as a fuel source during RE suggests a reduction of 25%–40% compared with pre-exercise glycogen levels (36). What is more intriguing are the kinetics of glycogen usage during RE (36), with the rate of use being associated with the intensity of the exercise performed (e.g., repetitions performed at a greater percentage of 1RM), whereas absolute glycogen usage is dependent on total workload (e.g., combination of number of repetitions, sets, and exercises at a given intensity) (36). This work clearly highlights that glycogen becomes an important metabolite during resistance exercise and that the intensity at which exercise is performed and the total workload are important modifiers of metabolic flux and throughput.

As previously highlighted, during glycolysis, phosphoglycerate dehydrogenase oxidizes 3-phosphoglycerate, creating the amino acid serine. Serine serves the one-carbon pool, which actively recycles homocysteine into methionine, giving rise to greater bioavailability of the precursor for *de novo* methylation. It is, therefore, plausible to suggest that significant up-regulation in glycolysis observed during high-intensity aerobic and resistance-based exercise increases the production of serine that serves to increase methylation potential in skeletal muscle. In partial support of such a hypothesis, it has been shown that glycolysis regulates the expression of phosphoglycerate dehydrogenase as the key enzyme for serine biosynthesis in an exercise training mouse model analyzing cardiac muscle (37). The product of glycolysis, pyruvate, undergoes a three-step, carbon degenerative process in which a resultant acetyl group is transferred to CoA resulting in acetyl-CoA, as previously described. Notably, however, under periods of oxygen restriction (i.e., acute, intense, and repetitive exercise bouts such as sprinting or RE), this process is not efficient enough to deliver acetyl-CoA to the mitochondria to meet the ATP demands of the cell. Thus, anaerobic glycolysis occurs to alleviate this demand, converting pyruvate into lactate via the enzyme lactate dehydrogenase (LDH) and supplying ATP to the working muscle in a reaction that occurs ~100 times that of aerobic metabolism, albeit only supplying 2 rather than 38 ATP molecules. Nonetheless, lactate production is synonymous with highly intense, repetitive, short bursts of exercise. For example, bilateral leg RE (in the form of squats) performed to exhaustion across two sets (70% and 50% 1RM to fatigue) increases blood lactate from resting levels of ~1.5 to >7.5 mmol (38).

Lactate as a metabolite has a distinct and direct role in cellular epigenetics. Indeed, in a series of intricate experiments, it has been shown, both *in vivo* and *in vitro*, that lactate is an endogenous HDAC inhibitor targeting the histone 4 complex, thereby acetylating histone marks, thus associating these modifications with increased gene expression (39). Indeed, via the use of gene expression arrays, the authors identified an association between lactate-induced histone acetylation modifications and gene expression. Interestingly, these authors also report that a cluster of genes whose expression is modified due to lactate treatment also are regulated after butyrate treatment (39). Butyrate is a well-known short-chain fatty acid (FA) that has significant HDAC inhibition characteristics, as described below. These data suggest that epigenetic dysregulation induced by lactate availability, a product of carbohydrate metabolism, shares common regulatory pathways with metabolites associated with FA metabolism (Fig. 3).

Typically, lactate has recently been implicated as a precursor substrate to the aptly named epigenetic modification, lactylation (40), further linking metabolism with the regulation of gene transcription. Indeed, Zhang *et al.* (40), first identified that lysine amino acid residues are susceptible to the addition of lactyl groups. Using stable carbon isotopes ($^{13}\text{C}_3$ -lactate), the authors elegantly identify that lactate itself is the precursor to this phenomenon and that such an epigenetic modification increases when cells were treated with rising levels of lactate. Functionally, lysine lactylation seems to occur predominantly in the promotor regions of coding genes and is positively correlated with the expression of these transcripts,

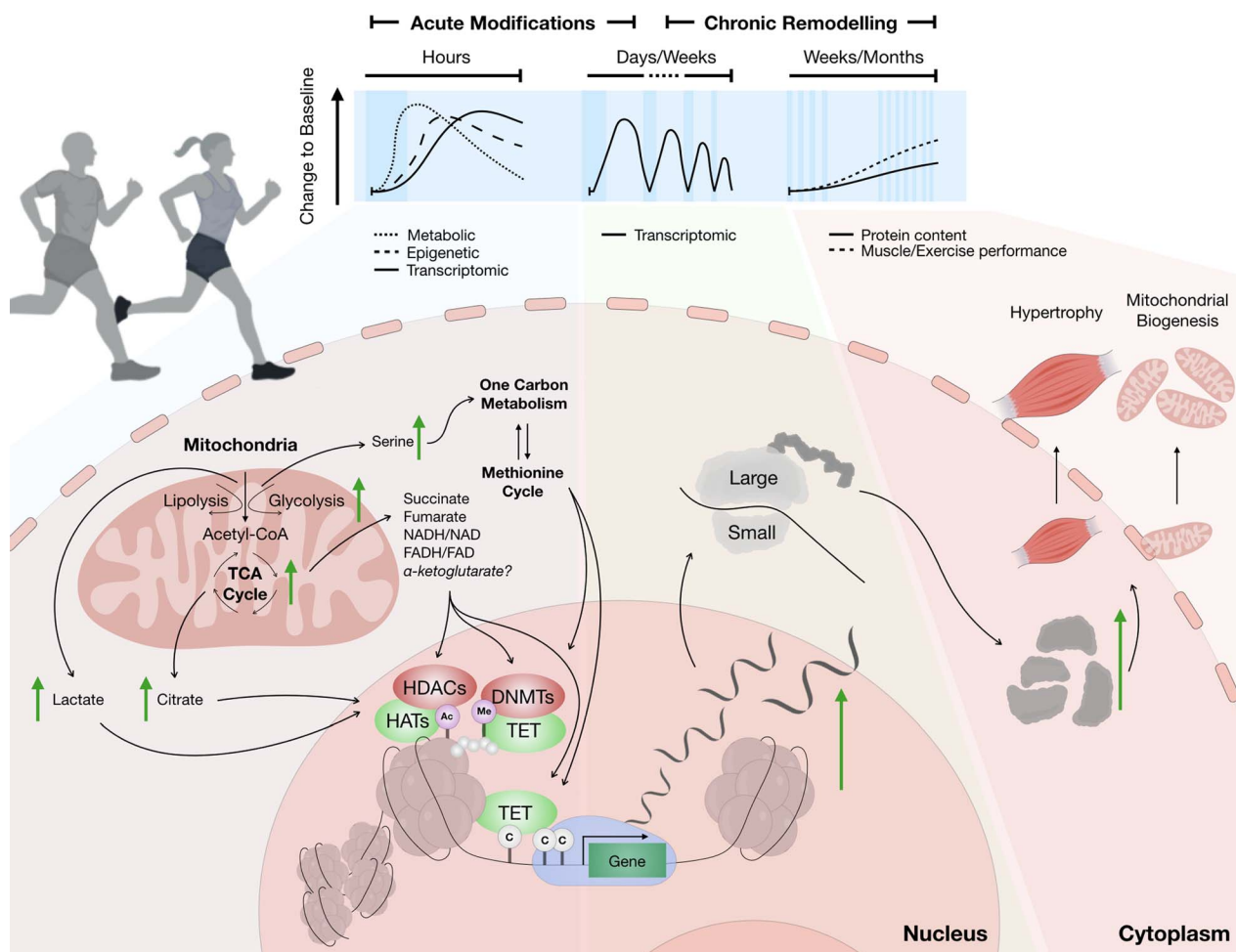


Figure 3. A schematic representation of the effect of exercise-induced metabolic dysregulation, epigenetic modifications, transcriptional consequences, and skeletal muscle adaptations. At the acute time frame (shaded in blue), exercise induces a transient modification in the bioavailability of metabolites, substrates, and cofactors, which in turn leads to fluctuations in the epigenetic state and subsequent transcriptional changes. Over proceeding days and weeks of repetitive exposure (shaded in green), exercise leads to fluctuations in transcriptomic behavior, resulting in both attenuated and augmented expression of candidate genes. These transcriptomic changes lead to functional consequences at the protein content level after weeks and months of repeated exercise stimuli (shaded in red), orchestrating phenotypic changes in skeletal muscle tissue and enhanced performance outcomes. Figure 3 can be viewed in color online at www.acsm-essr.org.

suggesting a protranscriptional characteristic of the modification (41). The mechanisms by which lysine lactylation occurs are as yet unknown, and further mechanistic work is needed. However, in an exercise perspective, this raises the tantalizing possibility that lactate production during and after intense exercise (as described above) directly regulates the epigenetic landscape of the genome and, by proxy, the transcriptional capacity of coding genes. Furthermore, because lactate is routinely shuttled between cells and tissues, this may also suggest that lactate production during/after exercise enables cross-talk between cells and tissues, facilitating broader adaptive responses within the biological milieu.

Importantly, this work implies that lactate is an important regulator of epigenetic enzyme activity, displaying a causal link between metabolic cell state, epigenetics, and cellular transcriptional potential. To date, however, no direct work has investigated the role of lactate in inducing epigenetic modifications in skeletal muscle after exercise stress, the resulting transcriptomic response, and functional remodeling of skeletal muscle, and thus, given the large-scale lactate production after exercise, this warrants further investigation.

Lipid Metabolism and Exercise

Lipolysis is the process of breaking down fats into its components of a glycerol backbone and FAs, with the latter serving several metabolic processes. FAs with aliphatic tails consisting of 12 or fewer carbon molecules (short-/medium-chain FAs) are able to diffuse across the mitochondrial membrane without assistance. Conversely, (very)long-chain FAs require transportation via the carnitine shuttle, where the carnitine palmitoyl transferase enzymes become rate limiting. FAs continue on their metabolic pathway by serving the β -oxidation pathway. The end product of β -oxidation is acetyl-CoA, which then enters the TCA cycle. In a secondary form of lipid-derived energy production, ketone bodies are formed in the liver during periods of metabolic stress, where acetyl-CoA and acetoacetyl-CoA interact, rendering acetoacetate (AcAc). AcAc is the primary ketone produced in the liver, where a minimal amount is excreted in the circulatory system, but with the majority being broken down by 3- β -hydroxybutyrate dehydrogenase (BDH) into β -hydroxybutyrate (β OHB). Monocarboxylate transports (MCT) in the sarcolemma and mitochondria are responsible for shuttling the ketone bodies to and from the circulating

systems and facilitating uptake into nonhepatic metabolically demanding tissues, such as skeletal muscle. The epigenetic implications of these metabolites have recently become of interest. In FA metabolism, butyrate, a short-chain FA, is produced and is known to have a positive effect on skeletal muscle health (42). Further, because of its kinetics has been reported to be a potent inhibitor of HDAC activity (43) (Fig. 3). More interestingly, ketone bodies have been suggested to have important signaling properties, specifically highlighting the ketones AcAc and β OHB and their effect on histone acetylation. Indeed, *in vitro*, it has recently been shown that β OHB inhibits HDACs 1, 3, and 4, where cells treated with a biologically relevant dose (e.g., 2–5 mM) of the ketone body exhibit a dose-dependent hyperacetylation of marks H3K9 and 14 (44). Similar effects are observed *in vivo*, where osmotic pump infusion of β OHB into the kidney of mice evokes histone hyperacetylation that consequently up-regulates the expression of specific transcripts, including the induction of forkhead box O3 (Foxo3) (44). In an exercise-related context, classical research shows that after prolonged (>90 min) intense exercise, either cycling or treadmill running has been shown to produce a significant increase (1–2 mM) in circulating β OHB levels in humans (45). Furthermore, this work highlighted that during exercise, ketone body uptake in skeletal muscle is elevated as much as fivefold that of basal, homeostatic levels (46). Given that β OHB constitutes the most abundant supply of ketone bodies within the circulating system, these data suggest a significant increase in production and uptake of β OHB during prolonged endurance exercise. Endurance exercise, at intensities of greater than ~65% maximal workload, relies heavily on carbohydrate-derived sources of energy (47), but with prolonged exertion, a greater reliance on lipid-derived energy is sought (48). These findings support the notion that prolonged endurance exercise increases not only the supply and uptake of β OHB but also the availability of free fatty acids (FFAs), such as butyrate. Given the known role β OHB and butyrate play in inhibiting HDACs, and the consequential effects on H3K9/27ac, we suggest that metabolic alterations induced by endurance exercise may promote epigenetic modifications, which have transcriptionally relevant consequences (Fig. 3). To date, less is known about FA metabolism during resistance exercise training in the context of β OHB availability and linking this to epigenetic modifications. Future work exploring such a proposition is, therefore, required.

Acetyl-CoA and the TCA Cycle

The overarching aim of metabolism is to provide energy to the meet the demands of the working skeletal muscle cell, a biological phenomenon that, upon exercise, must dramatically increase its throughput. The two key components of increasing the quantity of acetyl-CoA are the pyruvate dehydrogenase (PDH) complex for glucose metabolism and the β -oxidation cycle for metabolism of FAs. However, it is generally thought that the former pathway (PDH) supersedes that of the β -oxidation pathway and provides a greater amount of acetyl-CoA to the working muscle. Therefore, we shall discuss the role of glycolytic derived acetyl-CoA and its effects on TCA cycle throughput, intermediates, and the epigenetic effects.

PDH catalyzes the decarboxylation of pyruvate to acetyl-CoA during periods of high glucose metabolism. The activity of PDH itself is regulated by its phosphorylation status, in which PDH

kinases (PDK) phosphorylate the enzyme, rendering it inactive, and a group of PDH phosphatases act reciprocally. The active form of PDH has been shown to significantly increase almost instantly following exercise, ranging from exercise intensities of 35% to 90% max power output (49). However, when exercise sessions were prolonged to 10 min, the activity of PDH further increased at exercise intensities of 90%, whereas no further increase was observed at lower intensities. Collectively, this seminal work (49) identified that upon acute aerobic exercise, at ranging intensities, PDH is rapidly and significantly enhanced in terms of its activity. Long-term aerobic training has also been shown to increase PDH activity, where 8 wk of cycle ergometer training at 75% $\dot{V}O_{2\max}$ increased total PDH levels by ~30% in human skeletal muscle (50). It was also shown that PDK activity significantly increased after the 8-wk training regimen, which suggested that skeletal muscle, after training, harbored a greater sensitivity to PDH activity requirements. Again, these data report an increase in PDH activity suggesting a significant increase in glucose-derived acetyl-CoA. But also, the efficiency in triggering the active state of PDH, and thus the ability to efficiently regulate and control acetyl-CoA production is enhanced after long-term aerobic training due to increased activity of PDK. Once derived, acetyl-CoA enters the TCA cycle and undergoes a chain of reactions that incorporate and derive a host of aforementioned intermediates, some of which have distinct roles in regulating the activity of epigenetic enzymes (Fig. 3). In accordance with a large increase in acetyl-CoA availability, the TCA cycle undergoes an increase in productivity during aerobic-based exercise at both submaximal and supramaximal intensities. For example, after 10 min of steady-state (~60% of maximal power output for 10 min) unilateral knee extensor exercise, the total intramuscular concentration of the TCA cycle intermediates in recreationally active human skeletal muscle increased from ~1.4 to ~2.9 mmol kg⁻¹ dry weight (51). In this experiment, the same participants also showed a significant increase in total TCA intermediate concentration after exhaustive exercise, where concentrations rose from baseline to ~5.4 mmol kg⁻¹ dry weight upon volitional exhaustion (51). Furthermore, the same study showed an approximate 70-fold increase in TCA cycle flux during submaximal exercise and a near 100-fold increase after exhaustive exercise (51). Specifically, work identifies that the key intermediates; citrate, succinate, and fumarate, were all significantly up-regulated during both acute and more chronic aerobic exercise. Indeed, the aforementioned study showed a twofold increase in citrate, succinate, and fumarate after 5 min of steady-state knee extensor exercise (51) and, interestingly, showed a two- and threefold increase in both succinate and fumarate after their exhaustive knee extensor protocol, respectively (52). In a separate study, steady-state prolonged exercise (75% $\dot{V}O_{2\max}$, cycle ergometer) until fatigue (~75 min) showed a significant increase in both fumarate and citrate after only 5 min of exercise (52). With the latter metabolite further increasing after 40 min of exercise, before plateauing to volitional fatigue (52). The effects of exercise on levels of α -KG are less conclusive, with discordant findings showing both increases after acute and more prolonged aerobic exercise (53), and others showing little to no change (51), and, thus, conclusive work must be performed to understand the kinetics of α -KG during and after differing exercise

modalities and the effect this plays on the skeletal muscle epigenome.

The most conclusive evidence of a link between exercise-induced modifications to TCA cycle intermediates having an effect on epigenetic modifications and subsequent skeletal muscle remodeling comes from citrate (Fig. 3). The nuclear form of acetyl-CoA is known to be the pivotal contributor of acetyl molecules in the induction of histone acetylation. To this end, mitochondrial derived citrate serves to supply nuclear acetyl-CoA via a reaction with ACLY, which takes citrate through the cytoplasm into the nucleus, deriving acetyl-CoA and oxaloacetate. To this end, ACLY is a pivotal enzyme in the interplay between cellular metabolic state and epigenetic modifications at the histone level and, therefore, becomes, at least in part, a rate limiting step in the transcriptional competency of the cell. In a skeletal muscle context, ACLY has recently been implicated in both myoblast and satellite cell differentiation. Indeed, an ACLY silencing model (54) elegantly showed that in human and mouse myoblasts, differentiation is impaired, which is attributed to a decrease in gene expression of a number of myosin heavy chain isoforms, as well as a reduction in the key myogenic regulatory factor, MyoD (54). Conversely, they demonstrated that overexpression of ACLY increases MyoD expression due to ACLY-induced hyperacetylation of K9, 14, and 27 on histone protein 3, which congregate within the promotor region of the MyoD transcript. The authors finally identified that ACLY lay downstream of, and became a target for, insulin-like growth factor I (IGF-I), which is classically known to have important roles in muscle cell proliferation, differentiation, and survival. Collectively, this work suggests an important role for the enzyme ACLY in regulating skeletal muscle adaptability and myogenic transcriptional responses, in part due to its role in regulating H3K9, 14, and 27 acetylation levels. In this regard, it is plausible that exercise-induced systemic and local (to muscle) induction of IGF-I, which occurs after resistance exercise, may stimulate the pathway leading to the induction of ACLY and the conversion of citrate to acetyl-CoA. It is possible that a combination of increased metabolic flux, deriving greater quantities of nuclear citrate, is met with an IGF-I induction of ACLY, which collectively increases nuclear acetyl-CoA, permitting hyperacetylation of key skeletal muscle remodeling transcripts. Such an intracellular corroborative network has not been directly been investigated, but with such suggestive evidence, explorative work in the future will be important to confirm these hypotheses. Interestingly, recent work has suggested that glucose-derived and FA-derived acetyl-CoA molecules alter histone acetylation in two distinct mechanisms, with the latter mechanism utilizing acetate as a mediator (55). This work raises questions as to whether pre-exercise diet, substrate availability during exercise, and postexercise nutritional regimen may enhance epigenetic modifications in human skeletal muscle.

Metabolic Signaling Intermediates, Exercise, and Epigenetics

Biologically, there are several important molecules that act in the skeletal muscle milieu to generate muscular contraction that may have important epigenetic characteristics, the most prominent, and herein discussed, is calcium. Calcium ions (Ca^{2+}) play a pivotal role in skeletal muscle contraction, facilitating the

binding of the myosin head to the actin receptor, where, during exercise, myofibrillar Ca^{2+} levels increase by up to 100-fold (56). Downstream signaling effects of Ca^{2+} occur via the association of the ion with calmodulin, forming a complex (CaM) that subsequently targets several important enzymes. Of these targets is the nuclear Ca^{2+} /calmodulin-dependent protein kinase II (CaMKII), a serine/threonine-specific protein kinase that upon sufficient levels of Ca^{2+} signaling becomes phosphorylated, activated, and has the ability to autoregulate this active state (57). Activated nuclear CaMKII phosphorylates methyl CpG binding protein 2 (MeCP2), a protein that is known to act as a global chromatin modifier through association with gene promotor HDAC complexes and histone tail methylation states (58), thereby attenuating transcriptional potential. Recently, it was shown that CaMKII signaling is required for the phosphorylation of MeCP2, disrupting the localization of the regulatory factor away from the DNA and thus permitting transcriptional activity in neuronal cells (59). To the authors' knowledge, no work has, as yet, explored a link between acute high-intensity exercise metabolite signaling and MeCP2 modifications; however, partial support for this has been shown previously. Indeed, acute high-intensity exercise *in vivo* was shown to induce CaMKII phosphorylation through threonine-287 (60). Interestingly, in human resistance exercise-trained subjects, our group has recently identified CaMKIV as being a target of DNA methylation modifications (61). Indeed, after 7 wk of chronic resistance training, we showed that CaMKIV displayed a reduction in DNA methylation that persisted during a further 7 wk of detraining and a secondary period of training (7 wk of retraining). In this model, we demonstrated that reduction in DNA methylation of CaMKIV is concomitant with increases in the transcript at the expression level, suggesting an association between epigenetics and gene expression in exercising muscle (61). These data may suggest an intriguing epigenetic feedback mechanism, whereby enzymatic modifications induce self-regulation of the signaling pathway (Fig. 3).

In a second mode of action, Ca^{2+} and CaM signaling instigates the phosphorylation of the transcriptionally repressive epigenetic enzymes, HDAC4 and HDAC5. Indeed, in cardiac myocytes (62), it has been shown that local nuclear envelope Ca^{2+} release activates CaMKII, phosphorylating HDAC5 that subsequently translocates out of the nucleus, attenuating its transcriptionally repressive capacity (62). Via a unique docking site, CaMKII also targets, and HDAC4 initiates, phosphorylation of serine-246, serine-467, and serine-632, which not only increases its translocation but also represses the importation of the deacetylase enzyme (63). The role of Ca^{2+} to HDAC signaling cascade has been shown to be of significant importance in muscle remodeling, where CaMKII activation is sufficient to promote myogenesis and a hypertrophic response in cardiac myocytes (64). In human skeletal muscle, 40 min of cycling (at $\sim 75\% \dot{V}\text{O}_{2\text{peak}}$) (65) has been shown to induce a 50%–70% increase in autophosphorylation of CaMKII at threonine-278 (65), suggesting a link between more prolonged contractile activity and Ca^{2+} -induced muscular remodeling. A link between Ca^{2+} availability and a cascade of events leading to epigenetic modifications is therefore apparent, albeit with evidence being derived from fields external to that of exercise metabolism. However, there is strong evidence in muscle tissue (both cardiac and

skeletal) that Ca²⁺/calmodulin regulates the phosphorylated state of CaMKII, leading to the suggestion that epigenetic modifications may be also likely to occur in skeletal muscle tissue. It is, therefore, plausible to suggest that the exercise-induced up-regulation of Ca²⁺ dynamics within the muscle leads to a series of intracellular and intranuclear events, subsequent epigenetic modifications, and eventual tissue remodeling.

It is now becoming increasingly evident that exercise is a stressor to skeletal muscle, which is sufficient to induce widespread changes to DNA and histone epigenetic marks. Despite recent growth within the field however, a distinct paucity remains in our fundamental understanding of exercise-induced epigenetic modifications in skeletal muscle. Given our understanding of the interplay between exercise metabolism, epigenetics, and skeletal muscle remodeling in this review, tantalizing future research awaits molecular biologists/physiologists. Indeed, it is plausible that metabolically induced epigenetic modifications play a distinct regulatory role in physiological phenomena. For example, are epigenetic modifications enhanced when aerobic exercise is performed under deliberately attenuated glycogen/nutrient availability, explaining the augmented transcriptional response under these conditions? Or, do circulating metabolites induce epigenetic modifications in nonexercised limbs during RT programs, explaining the classically observed systemic effect of exercise? Such research questions await to be elucidated. What is clear, however, is that skeletal muscle epigenetics promises to be an exciting, fascinating, and prosperous field of biological research for decades to come.

CONCLUDING REMARKS

To date, exercise, coupled with adequate nutritional regimen, remains the foremost intervention in maintaining health, as well as ameliorating disease states. Understanding of the pleiotropic mechanism by which exercise elicits these effects suggests a complex and integrated network of signaling pathways that orchestrate functional consequences at a whole-tissue/morphological level. Here, we summarize our current understanding of the interplay between exercise-mediated metabolic regulation, the induction of epigenetic modifying enzymes, epigenetic modifications, and the co-ordinated transcriptional outcome. This review therefore suggests epigenetics to be the key link between exercise and transcriptional/functional remodeling of skeletal muscle. Despite the work required to solidify our understanding of this area, the interplay between exercise and epigenetics is becoming well established and, thus, can no longer be ignored in the field of exercise biology. Moving forward, therefore, it is of crucial importance for molecular exercise biologists to pursue lines of research that progress our understanding of this interplay to facilitate better development of interventions and therapeutics of skeletal muscle disease.

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